

AI in Industry

An Oddball

This is a strange course

For most courses, it's easy to figure out what to expect:

- "Introduction to Algorithms and Programming"
- "Statistical and Mathematical Methods for Artificial Intelligence"
- "Machine Learning"
- "Deep Learning"
- "Combinatorial Decision Making and Optimization"
- ...

...But what for something called "AI in Industry"?

What do we mean by "industry"?

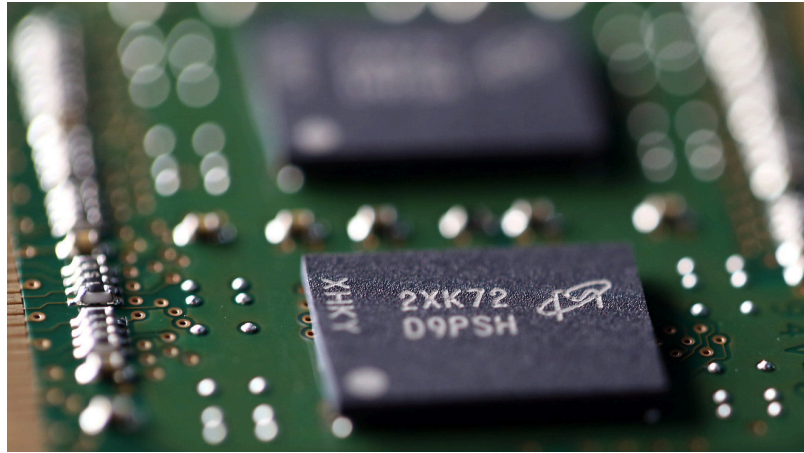
Industry

This is industry



Industry

This is industry



Industry

This is also industry



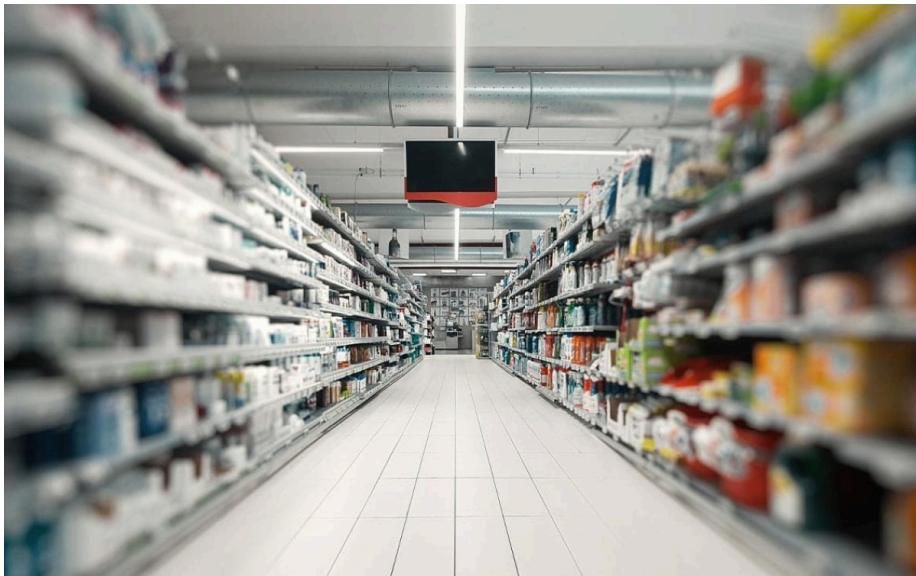
Industry

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Industry

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Industry

We are talking about industry in a very *broad* sense! Because...

We will define "industry" as any activity that can generate *value*

- So, this course is about how to *using AI to generate value*
- ...And since value is typically generated by solving problems

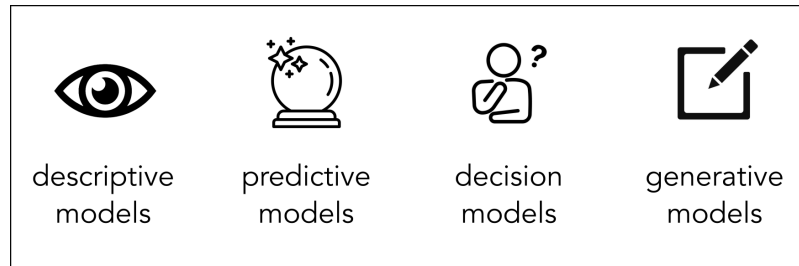
Sooo...

This course is about using AI to *address real world problems*

Which problems are we talking about?

Models

A good starting point: AI is mainly about *models*



So we can, for example:

- Present data in an insightful fashion
- Make estimates or forecasts of unknown quantities or classes
- Recommend optimal or near-optimal decision
- Generate human-interpretable content

Order and Chaos

This is simple and useful characterization

...But the truth is more like this!



Order and Chaos

Industrial applications are *messy*

- The problems are not well defined
- Similar techniques may be applied in multiple settings
- ...And with different names
- Classical tasks do not cover whole problem
- We often need to combine problems/techniques
- ...

A common opinion:

try something, then add tweaks until the problem is solved

Order and Chaos

...But this is *evil*!



Order and Chaos

Specifically, it reaches a *plateau* real quick:

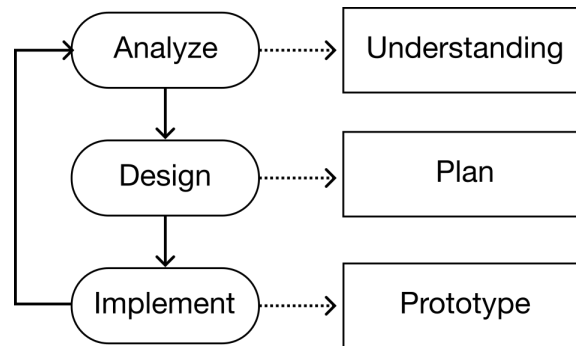
- If you get lucky, you solve your problem and you do it fast
- But more often than not:
 - You fail, and you *don't understand why*
 - You end up with a much *messier solution* than needed
 - Your approach works on test data, but *not in the field*

This can become *very frustrating* (at best)

- Instead, we will try to bring a *measure of order* to the chaos
- ...Which is actually impossible, but still the right choice on the long run

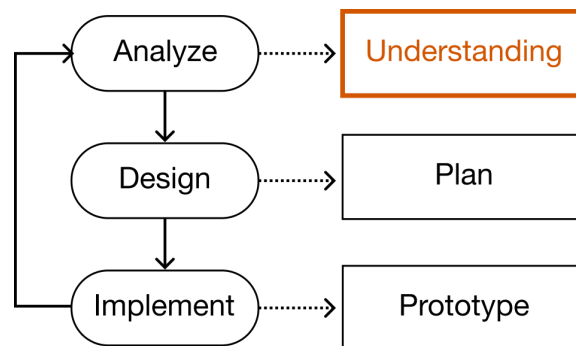
A Better Alternative

Without getting to philosophical, here's a much better process



A Better Alternative

Without getting to philosophical, here's a much better process



- The key point is *understanding*
- As long as you *understand better*, you can *get better*

What to Expect

How I am Going to Play It

I am going to follow a few guiding principles

How: Examples! I.e. Use Cases

- Every few lectures we will introduce a *new use case*
- They will be *simplified* industrial problems
 - Real industrial problems would take too much to tackle
 - ...Not to mention they are subject to NdAs 😞
- They will nevertheless be *representative*
- Some uses cases will be covered in *seminars* by industrial partners

How I am Going to Play It

I am going to follow a few guiding principles

What: techniques, best practices, formalization

- Mostly: how to *methodically tackle* a new problem
- But we will also introduce *new techniques*
- ...Ways to *apply* known techniques
- ...Ways to *combine* known techniques
- ...And how to *formalize* problems and ideas

How I am Going to Play It

I am going to follow a few guiding principles

Why: my goal is for you to tackle problems better than most of your peers

- Problems/solutions are often poorly understood
 - Formalizing is the first step towards understanding
- Different problems call for different tools
 - Using (say) ML for everything is just *inefficient*
- Many people can apply "boilerplate", mainstream AI methods
 - ...But much fewer are capable of *changing* or *combining* them

On the Art of Cooking

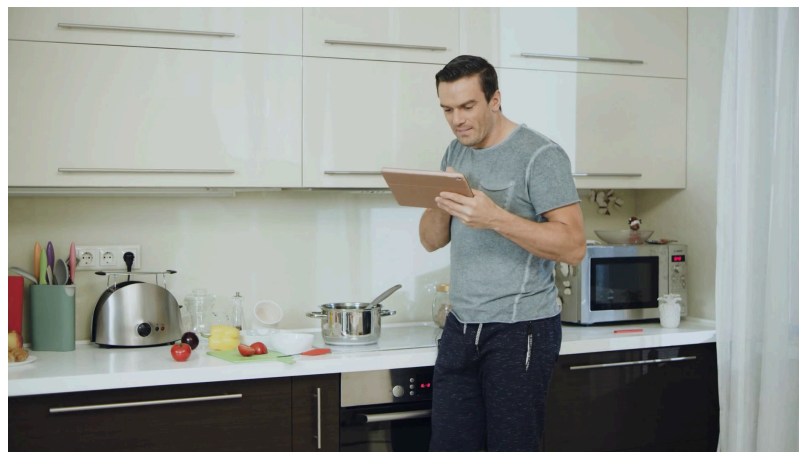
At some point, the course will start feeling like a cookbook



When you get there, there is *one thing you should remember*

On the Art of Cooking

Most people read cookbooks to follow recipes



On the Art of Cooking

...But *true chefs* read cookbooks to find *ideas*



So, learn, then *get creative!*

Technical Information

Teachers

Teacher:

- Michele Lombardi (michele.lombardi2@unibo.it)
- Office phone: 051 2093270 (it's close to teaching room 5.7)

Student hours: you can book an appointment (online by default) via:

<https://book.morgen.so/michelelombardi03/student-hours>

Tutor:

- TBA
- Main contact point when picking a project topic
- Student hours: TBA

Course Material

Reference: [course web site on virtuale.unibo.it](#)

- Jupyter notebooks + `requirements.txt` + [poetry configuration files](#)
- PDF notes (there are just meant as a backup solution)
- Recorded lectures (via Panopto, links on the web site)

This course changes (a bit) every year

- The good part: the course will *grow with you*
- The bad part: lecture material will typically arrive one/two days early at most

Exam

The exam will consist of *a project*:

- You can *propose a topic* (this is the most typical solution)
- ...Or *pick one* from the list on <https://lia-unibo.github.io/>
- The topic must be *discussed with the tutor and the teacher* before starting
- Groups of 2-3 students tend to work best
- ...But individual projects are also fine

An advice: *wait until at least mid course* before choosing

Once you are done with your work

...You'll need to schedule a call (online by default) via:

<https://book.morgen.so/michelelombardi03/a3i-exams>

Exam

The students will need to:

- Deliver the project code
- Give a presentation
- Be prepared to discuss their work (including a code check)

The evaluation

- Will *not* focus on how successful your results are
- ...But on *how you reached them*
- This means I will evaluate:
 - *Why* you made the choices you made
 - *How* you have interpreted the results
 - Your familiarity with the *techniques* you chose to employ

Generative AI Policy

Generative AI tools can be freely used

...Provided that you end up with *better understanding*

- Ask how, what and why
- Avoid delegation, except for things you already know
- Challenge what the LLM says, if you are not convinced

At the test, you are expected to own everything you present

- Make sure you can explain and justify your choices
- ...Your results
- ...and your code!

The Exam and the 3-ECTS Project

About the *optional 3-credit project*

- You can combine them
- The 3CFU project will be *a follow-up of the exam work*
- Typically, both works are presented at once
- ...But you can have separate presentations if you wish

If you wish to combine the projects

...You should mention that when you pick the topic

- We'll make sure that the topic is broad enough for both

The 3-ECTS project does not have an actual grade

- Either you pass, or you don't
- By the time you are ready to present, you'll typically be fine